LA-DT Framework: Appendix A — Proofs and Derivations

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1 Proofs and Derivations

This appendix provides complete proofs for the theoretical claims in the LA-DT paper. Section 1.1 restates the formal assumptions. Section 1.2 establishes preliminary lemmas on mean squared residuals. Section 1.3 proves the main VGR discrimination result (Theorem 1). Section 1.4 analyzes the DEV and SCD signals. Section 1.5 derives the LLR scoring rule. Section 1.6 proves that honest-majority violation renders attribution impossible (Theorem 2). Section 1.7 extends the discrimination guarantee to polynomial drift.

1.1 Notation and Assumptions

We adopt the standard notation from the main paper throughout. A sensor network $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ consists of $N = |\mathcal{V}|$ sensors. The set of Byzantine-compromised sensors is $\mathcal{B} \subseteq \mathcal{V}$ with $f = |\mathcal{B}|$. Each sensor i produces reading $y_i(t)$ at discrete timestep $t \in \{1, \ldots, T\}$ (at 1 Hz sampling). The digital twin produces prediction $\hat{y}_i(t)$, and the residual is $r_i(t) = y_i(t) - \hat{y}_i(t)$. The observation window of length T is split into two equal halves: the *first half* $[1, n]$ and the *second half* $[n{+}1, 2n]$, where $n = T/2$. The Mean Squared Residual (MSR) over an interval $[a, b]$ is:

$$\mathrm{MSR}(r_i)_{[a,b]} \;=\; \frac{1}{b-a+1} \sum_{t=a}^{b} r_i(t)^2. \tag{1}$$

The following assumptions are the five core axioms underlying the attribution framework:

Definition 1 (Attribution Assumptions).

A1 (Honest Majority). *The number of Byzantine sensors satisfies $f < N/2$.*

A2 (Linear Drift Model). *Each Byzantine sensor $i \in \mathcal{B}$ injects drift $\Delta y_i(t) = \delta_i \cdot t$ with rate $\delta_i > 0$.*

A3 (IID Natural Noise). *Measurement noise $\eta_i(t) \sim \mathcal{N}(0, \sigma_\eta^2)$ is independent across sensors and time.*

A4 (Variance Stationarity). *The noise variance σ_η^2 is constant over the observation window $[1, T]$.*

A5 (DT Fidelity). *The DT prediction error satisfies $|\hat{y}_i(t) - y_i^*(t)| < \epsilon_{\mathrm{DT}}$ with $\epsilon_{\mathrm{DT}} \ll \delta_i T$ for all honest sensors $i \notin \mathcal{B}$.*

Under A3 and A5, the residual for an honest sensor satisfies $r_i(t) = \eta_i(t) + O(\epsilon_{\mathrm{DT}})$, which we approximate as $r_i(t) \approx \eta_i(t)$. For a Byzantine sensor, $r_i(t) = \delta_i t + \eta_i(t) + O(\epsilon_{\mathrm{DT}}) \approx \delta_i t + \eta_i(t)$.

1.2 Preliminary Lemmas

Lemma 1 (MSR of Stationary White Noise). *Let $\{r_i(t)\}_{t=1}^n$ be i.i.d. $\mathcal{N}(0, \sigma^2)$. Define $\mathrm{MSR} = \frac{1}{n}\sum_{t=1}^n r_i(t)^2$. Then:*

(a) $E[\mathrm{MSR}] = \sigma^2$.
(b) $\mathrm{Var}(\mathrm{MSR}) = 2\sigma^4/n$.
(c) For any $\epsilon \in (0, 2\sigma^2]$:

$$P\big(|\mathrm{MSR} - \sigma^2| > \epsilon\big) \;\le\; 2\exp\left(-\frac{n\epsilon^2}{8\sigma^4}\right). \tag{2}$$

Proof. Parts (a) and (b) follow from $E[\eta^2] = \sigma^2$ and $\mathrm{Var}(\eta^2) = E[\eta^4] - (E[\eta^2])^2 = 3\sigma^4 - \sigma^4 = 2\sigma^4$ for $\eta \sim \mathcal{N}(0, \sigma^2)$, applied to the average of n i.i.d. copies.

For part (c), the variables $Z_t = r_i(t)^2 - \sigma^2$ are i.i.d., zero-mean, and sub-exponential with parameters $(\nu, b) = (2\sigma^2, 4\sigma^2)$. By Bernstein's inequality for sub-exponential random variables [1]:

$$P\left(\left|\frac{1}{n}\sum_{t=1}^n Z_t\right| > \epsilon\right) \le 2\exp\left(-\frac{n}{2}\min\left(\frac{\epsilon^2}{\nu^2}, \frac{\epsilon}{b}\right)\right).$$

For $\epsilon \le 2\sigma^2$, the quadratic term dominates, giving the stated bound. □

Lemma 2 (MSR Under Linear Drift). *Let $r_i(t) = \delta t + \eta_i(t)$ where $\eta_i(t) \sim \mathcal{N}(0, \sigma^2)$ are i.i.d. Define $n = T/2$ and let MSR_1, MSR_2 be the MSR computed over the first half $[1, n]$ and second half $[n{+}1, 2n]$. Then:*

(a) $E[\mathrm{MSR}_1] = \dfrac{\delta^2\, n(n+1)(2n+1)}{6n} + \sigma^2 = \dfrac{\delta^2(n+1)(2n+1)}{6} + \sigma^2$.
(b) $E[\mathrm{MSR}_2] = \dfrac{\delta^2(2n+1)(7n+1)}{6} + \sigma^2$.
(c) For large n:

$$E[\mathrm{MSR}_1] \;\approx\; \frac{\delta^2 T^2}{12} + \sigma^2, \qquad E[\mathrm{MSR}_2] \;\approx\; \frac{7\,\delta^2 T^2}{12} + \sigma^2. \tag{3}$$

Proof. Since $\eta_i(t) \perp\!\!\!\perp \delta t$ and $E[\eta_i(t)] = 0$:

$$E[r_i(t)^2] = E[(\delta t + \eta_i(t))^2] = \delta^2 t^2 + 2\delta t \cdot \underbrace{E[\eta_i(t)]}_{=0} + \underbrace{E[\eta_i(t)^2]}_{=\sigma^2} = \delta^2 t^2 + \sigma^2.$$

First half:

$$E[\mathrm{MSR}_1] = \frac{1}{n}\sum_{t=1}^n (\delta^2 t^2 + \sigma^2) = \frac{\delta^2}{n} \cdot \frac{n(n+1)(2n+1)}{6} + \sigma^2 = \frac{\delta^2(n+1)(2n+1)}{6} + \sigma^2.$$

For large n: $(n+1)(2n+1)/6 \approx 2n^2/6 = n^2/3 = T^2/12$, confirming part (c).

Second half:

$$\begin{aligned}
E[\mathrm{MSR}_2] &= \frac{1}{n}\sum_{t=n+1}^{2n}(\delta^2 t^2+\sigma^2) = \frac{\delta^2}{n}\left[\sum_{t=1}^{2n} t^2 - \sum_{t=1}^{n} t^2\right] + \sigma^2 \\
&= \frac{\delta^2}{n}\left[\frac{2n(2n+1)(4n+1)}{6} - \frac{n(n+1)(2n+1)}{6}\right] + \sigma^2 \\
&= \frac{\delta^2(2n+1)}{6n}\big[2n(4n+1) - n(n+1)\big] + \sigma^2 \\
&= \frac{\delta^2(2n+1)}{6n}\cdot n(7n+1) + \sigma^2 = \frac{\delta^2(2n+1)(7n+1)}{6} + \sigma^2. \qquad (4)
\end{aligned}$$

For large n: $(2n+1)(7n+1)/6 \approx 14n^2/6 = 7n^2/3 = 7T^2/12$, establishing part (c).

The ratio of the second-half to first-half summations (ignoring noise) is:

$$\frac{(2n+1)(7n+1)}{(n+1)(2n+1)} = \frac{7n+1}{n+1} \xrightarrow{n\to\infty} 7. \qquad (5)$$

This is the origin of the coefficient $k = 7$ shown in Eq. (5). □

Remark 1. The coefficient $k = 7$ has a concise interpretation via continuous integration: $\int_{T/2}^{T} t^2\,dt / \int_0^{T/2} t^2\,dt = (T^3 - T^3/8)/(T^3/8) = 7$. The discrete sum converges to this ratio as $n \to \infty$.

1.3 Theorem 1: VGR-Based Byzantine Discrimination

The central theoretical result of the LA-DT framework establishes that the Variance Growth Rate (VGR), defined as

$$\mathrm{VGR}_i = \frac{\mathrm{MSR}(r_i)_{[n+1,2n]}}{\mathrm{MSR}(r_i)_{[1,n]}}, \qquad (6)$$

provably separates honest and Byzantine sensors under assumptions A1–A5.

Theorem 1 (VGR-Based Byzantine Discrimination). *Let $N \geq 5$ sensors be observed over $T = 2n$ timesteps at 1 Hz, where each sensor i produces residuals $r_i(t) = y_i(t) - \hat{y}_i(t)$. Under assumptions A1–A5 (Definition 1):*

(i) ***Honest sensor bound.*** *For any honest sensor $i \notin \mathcal{B}$:*

$$P\big(\mathrm{VGR}_i \in [\,0.85,\ 1.15\,]\big) \ \geq\ 1 - 2\exp\left(-\frac{0.004\,n}{\sigma_\eta^4}\cdot\sigma_\eta^4\right) = 1 - 2e^{-0.004\,n}. \qquad (7)$$

In particular, for observation windows with $n \geq 900$ (i.e., $T \geq 1800$ timesteps, corresponding to $H \geq 30$ min at 1 Hz), this probability exceeds 0.97.

(ii) ***Byzantine sensor divergence.*** *For any Byzantine sensor $i \in \mathcal{B}$ with drift rate $\delta_i > 0$, the expected VGR satisfies:*

$$E[\mathrm{VGR}_i] = \frac{7\,\delta_i^2 T^2/12 + \sigma_\eta^2}{\delta_i^2 T^2/12 + \sigma_\eta^2}. \tag{8}$$

In the high-SNR regime ($\delta_i^2 T^2 \gg 12\,\sigma_\eta^2$), $E[\mathrm{VGR}_i] \to 7$.

(iii) ***Separation guarantee.*** *Defining $\mathrm{SNR}_i = \delta_i^2 T^2/(12\,\sigma_\eta^2)$, the expected VGR exceeds 3.0 whenever $\mathrm{SNR}_i \geq 0.5$, i.e., $\delta_i T \geq \sqrt{6}\,\sigma_\eta$. For typical parameters ($\delta_i = 0.01$, $T = 3600$, $\sigma_\eta = 0.1$), $\mathrm{SNR}_i = 2700$, yielding a gap of $E[\mathrm{VGR}_{\mathrm{byz}}]/E[\mathrm{VGR}_{\mathrm{hon}}] \approx 7.0$.*

Proof. Part (i): Honest sensor. Under A3–A5, an honest sensor has $r_i(t) = \eta_i(t) \sim \mathcal{N}(0, \sigma_\eta^2)$ i.i.d. By Lemma 1: $E[\mathrm{MSR}_1] = E[\mathrm{MSR}_2] = \sigma_\eta^2$. The ratio $\mathrm{VGR}_i = \mathrm{MSR}_2/\mathrm{MSR}_1$ follows an $F(n, n)$ distribution (ratio of two independent χ_n^2/n variables).

For large n, the log-F distribution is approximately Gaussian with variance $2/n$, so $\log(\mathrm{VGR}_i) \dot{\sim} \mathcal{N}(0,\ 2/n)$ [2]. Therefore:

$$\begin{aligned} P\big(\mathrm{VGR}_i \in [0.85, 1.15]\big) &= P\big(\log(0.85) \leq \log(\mathrm{VGR}_i) \leq \log(1.15)\big) \\ &= P\left(\frac{-0.163}{\sqrt{2/n}} \leq Z \leq \frac{0.140}{\sqrt{2/n}}\right) \end{aligned} \tag{9}$$

where $Z \sim \mathcal{N}(0, 1)$. For $n = 900$ ($H = 30$ min): $\sqrt{2/n} = 0.0471$, giving $P(-3.45 \leq Z \leq 2.97) \geq 0.998 > 0.97$. For $n = 1800$ ($H = 60$ min): $\sqrt{2/n} = 0.0333$, giving $P(-4.88 \leq Z \leq 4.20) > 0.9999$.

Eq. (7) follows from the Gaussian tail bound $P(|Z| > z) \leq 2e^{-z^2/2}$ applied with $z = 0.085\sqrt{n/2}$, using $\min(|\log 0.85|, \log 1.15) \geq 0.085\sqrt{2}$.

Part (ii): Byzantine sensor. Under A2, a Byzantine sensor has $r_i(t) = \delta_i t + \eta_i(t)$. By Lemma 2:

$$E[\mathrm{VGR}_i] = \frac{E[\mathrm{MSR}_2]}{E[\mathrm{MSR}_1]} \approx \frac{7\delta_i^2 T^2/12 + \sigma_\eta^2}{\delta_i^2 T^2/12 + \sigma_\eta^2} = \frac{7\,\mathrm{SNR}_i + 1}{\mathrm{SNR}_i + 1}.$$

As $\mathrm{SNR}_i \to \infty$, this ratio converges to 7.

This is a first-order approximation: $E[\mathrm{MSR}_2/\mathrm{MSR}_1] \neq E[\mathrm{MSR}_2]/E[\mathrm{MSR}_1]$ in general. However, since both MSR_1 and MSR_2 concentrate tightly around their means for large n (Lemma 1 applied to the noise component, with the deterministic drift contributing a fixed term), the ratio concentrates around $E[\mathrm{MSR}_2]/E[\mathrm{MSR}_1]$ by the continuous mapping theorem.

Part (iii): Separation. Setting $E[\mathrm{VGR}_i] \geq 3$:

$$\frac{7\,\mathrm{SNR}_i + 1}{\mathrm{SNR}_i + 1} \geq 3 \iff 7\,\mathrm{SNR}_i + 1 \geq 3\,\mathrm{SNR}_i + 3 \iff \mathrm{SNR}_i \geq \tfrac{1}{2}.$$

Since $\mathrm{SNR}_i = \delta_i^2 T^2/(12\sigma_\eta^2)$, the condition $\mathrm{SNR}_i \geq 1/2$ is equivalent to $\delta_i T \geq \sqrt{6}\,\sigma_\eta \approx 2.45\,\sigma_\eta$. For the experimental parameters ($\delta_i = 0.01$, $T = 3600$, $\sigma_\eta = 0.1$): $\delta_i T/\sigma_\eta = 360 \gg 2.45$, confirming separation with overwhelming margin. Combined with part (i), the gap between honest VGR $\in [0.85, 1.15]$ and Byzantine VGR ≥ 3.0 is robust, establishing discrimination with confidence ≥ 0.97 for $H \geq 30$ min. □

1.4 Analysis of DEV and SCD Signals

Theorem 1 establishes VGR-based separation. The following propositions show that DEV and SCD provide complementary discrimination, justifying their inclusion in the LLR-based scoring (Section 1.5).

Proposition 1 (DEV Divergence Under Byzantine Drift). *Under assumptions A1–A5, let $\tilde{r}(t) = \mathrm{median}_j(r_j(t))$ denote the spatial median residual. Then for n sufficiently large:*

(i) *For any honest sensor $i \notin \mathcal{B}$: $E[\mathrm{DEV}_i] \approx 1$.*
(ii) *For any Byzantine sensor $i \in \mathcal{B}$: $E[\mathrm{DEV}_i] \geq \delta_i T/(\delta_i T/2 + \epsilon) \to 2$ as $\delta_i T \to \infty$.*

Proof. Under A1 (honest majority), $\tilde{r}(t)$ is determined by honest sensors with high probability. By A3, honest residuals are i.i.d. $\mathcal{N}(0, \sigma_\eta^2)$, so $\tilde{r}(t) \approx 0$ for all t (the median of a symmetric distribution centered at 0).

Part (i): For honest sensor i, $r_i(t) - \tilde{r}(t) \approx \eta_i(t)$. Since $\eta_i(t)$ is stationary, the distribution of $\max_{t\in[1,n]} |\eta_i(t)|$ and $\max_{t\in[n+1,2n]} |\eta_i(t)|$ are identical. Hence the ratio $\mathrm{DEV}_i \approx 1$ in expectation.

Part (ii): For Byzantine sensor i, $r_i(t) - \tilde{r}(t) \approx \delta_i t + \eta_i(t)$. When drift dominates noise ($|\delta_i t| \gg |\eta_i(t)|$):

$$\max_{t\in[n+1,2n]} |r_i(t) - \tilde{r}(t)| \approx |\delta_i| \cdot 2n = |\delta_i| T,$$

$$\max_{t\in[1,n]} |r_i(t) - \tilde{r}(t)| \approx |\delta_i| \cdot n = |\delta_i| T/2.$$

The Deviation Ratio (DEV) is defined as the ratio of maximum deviations in the second temporal half to the first half:

$$\mathrm{DEV}_i = \frac{\max_{t\in[n+1,2n]} |r_i(t) - \tilde{r}(t)|}{\max_{t\in[1,n]} |r_i(t) - \tilde{r}(t)|}.$$

Substituting the above analysis: $\mathrm{DEV}_i \approx \delta_i T/(\delta_i T/2 + \epsilon) \to 2$. More generally, $\mathrm{DEV}_i > 1$ whenever drift grows faster than noise fluctuations, providing a monotonically increasing discrimination signal. □

Proposition 2 (SCD Under Byzantine Compromise). *Under assumptions A1–A5, for a Byzantine sensor $i \in \mathcal{B}$ and any honest sensor $j \notin \mathcal{B}$:*

(i) *The Pearson correlation* $\rho_{ij}^{[a,b]}$ *between* r_i *and* r_j *over interval* $[a,b]$ *satisfies* $\rho_{ij}^{[a,b]} \to 0$ *as* $\delta_i^2(b-a)^2 \to \infty$.

(ii) $\mathrm{SCD}_i(H) > 0$ *for sufficiently large* T*, since* $\rho_{ij}^{[1,n]} > \rho_{ij}^{[n+1,2n]}$ *when drift accumulates.*

Proof. The residuals are $r_i(t) = \delta_i t + \eta_i(t)$ (Byzantine) and $r_j(t) = \eta_j(t)$ (honest). Since $\eta_i \perp\!\!\!\perp \eta_j$ (A3):

$$\mathrm{Cov}(r_i(t), r_j(t)) = \mathrm{Cov}(\delta_i t + \eta_i(t),\ \eta_j(t)) = 0.$$

The Pearson correlation over $[a,b]$ is:

$$\rho_{ij}^{[a,b]} = \frac{\sum_{t=a}^{b}(r_i(t)-\bar{r}_i)(r_j(t)-\bar{r}_j)}{\sqrt{\sum_t (r_i(t)-\bar{r}_i)^2 \cdot \sum_t (r_j(t)-\bar{r}_j)^2}}.$$

The numerator has expectation zero (independence), so $E[\rho_{ij}^{[a,b]}] \approx 0$ by the law of large numbers. However, the *sample* correlation over a finite window fluctuates around zero with standard deviation $\approx 1/\sqrt{b-a}$. The denominator grows with drift magnitude since $\mathrm{Var}_t(r_i) = \mathrm{Var}_t(\delta_i t + \eta_i) = \mathrm{Var}_t(\delta_i t) + \sigma_\eta^2$, which increases with the window span.

For the first temporal half $[1,n]$, drift is small and the correlation fluctuation is relatively large. For the second half $[n+1,2n]$, the large drift variance in the denominator suppresses correlation magnitude. Thus $|\rho_{ij}^{[n+1,2n]}| < |\rho_{ij}^{[1,n]}|$ in expectation, producing positive Sensor Correlation Decay (SCD), defined as:

$$\mathrm{SCD}_i(H) = \frac{1}{N-1}\sum_{j\neq i}\big(\rho_{ij}^{[1,n]} - \rho_{ij}^{[n+1,2n]}\big).$$

This quantity is positive when a sensor's correlations decay in the second half, a hallmark of Byzantine drift.

Between two honest sensors $i, j \notin \mathcal{B}$: both residuals are stationary noise, so ρ_{ij} has the same distribution in both halves, yielding $\mathrm{SCD}_i \approx 0$. □

1.5 LLR Scoring Derivation

This section derives the weighted linear form of the Log-Likelihood Ratio (LLR), which combines the three discrimination signals.

Proposition 3 (LLR Under Exponential Family Signals). *Let* $s_i = (s_i^{\mathrm{VGR}},\ s_i^{\mathrm{DEV}},\ s_i^{\mathrm{SCD}}) \in \mathbb{R}^3$ *be the signal vector for sensor* i*. Assume that under both the honest hypothesis* H_0 *and the Byzantine hypothesis* H_1*, each signal component* $s_i^{(j)}$ *follows independently a distribution from a one-parameter exponential family:*

$$p(s_i^{(j)} \mid H_k) = h(s_i^{(j)})\exp\big(\theta_k^{(j)} s_i^{(j)} - A(\theta_k^{(j)})\big), \quad k \in \{0,1\}. \tag{10}$$

Then the log-likelihood ratio decomposes as:

$$\log \frac{p(\mathbf{s}_i \mid H_1)}{p(\mathbf{s}_i \mid H_0)} = \sum_{j \in \{\text{VGR,DEV,SCD}\}} w_j \cdot s_i^{(j)} + C, \tag{11}$$

where $w_j = \theta_1^{(j)} - \theta_0^{(j)}$ *and* $C = \sum_j \left[A(\theta_0^{(j)}) - A(\theta_1^{(j)})\right]$ *is a constant absorbed into the threshold* τ*.*

Proof. Under conditional independence of the three signals:

$$\begin{aligned}\log \frac{p(\mathbf{s}_i \mid H_1)}{p(\mathbf{s}_i \mid H_0)} &= \sum_j \log \frac{p(s_i^{(j)} \mid H_1)}{p(s_i^{(j)} \mid H_0)} \\ &= \sum_j \left[(\theta_1^{(j)} - \theta_0^{(j)}) s_i^{(j)} - \left(A(\theta_1^{(j)}) - A(\theta_0^{(j)})\right)\right].\end{aligned}$$

The $h(\cdot)$ base measure terms cancel in the ratio. Defining $w_j = \theta_1^{(j)} - \theta_0^{(j)}$ and collecting the log-normalizer differences into C yields Eq. (11). In practice, the weights w_j are estimated via logistic regression on training data, which directly fits the log-odds model $\log[P(H_1)/P(H_0)] = \sum_j w_j s_i^{(j)} + b$, subsuming the constant C into the bias term b and the decision threshold τ. □

Remark 2. The conditional independence assumption is approximate: VGR, DEV, and SCD are all computed from overlapping residuals $r_i(t)$. Logistic regression remains a consistent estimator of the Bayes-optimal linear decision boundary even when signals are correlated [3], which is why the pipeline uses empirically calibrated weights rather than assuming exact independence.

1.6 Theorem 2: Honest Majority Necessity

Theorem 2 (Honest Majority is Necessary for Attribution). *If* $|\mathcal{B}| \geq \lceil N/2 \rceil$ *(majority compromised), then there exist two configurations—one with Byzantine set* $\mathcal{B}_A$ *and one with* $\mathcal{B}_B = \mathcal{V} \setminus \mathcal{B}_A$*—that produce statistically indistinguishable observations. Consequently, no attribution algorithm based solely on inter-sensor statistics can correctly identify* $\mathcal{B}$ *with probability exceeding* $1/2 + o(1)$*.*

Proof. Consider N sensors partitioned into two groups: $G_1 = \{1, \ldots, m\}$ and $G_2 = \{m+1, \ldots, N\}$ with $m = \lceil N/2 \rceil$.

Scenario A: $\mathcal{B}_A = G_1$ (group 1 Byzantine). The true physical state is $x^*(t)$. Byzantine sensors inject drift: $y_i(t) = x^*(t) + \delta t + \eta_i(t)$ for $i \in G_1$. Honest sensors report: $y_j(t) = x^*(t) + \eta_j(t)$ for $j \in G_2$.

Scenario B: $\mathcal{B}_B = G_2$ (group 2 Byzantine). The true physical state is $\tilde{x}^*(t) = x^*(t) + \delta t$. Honest sensors report: $y_i(t) = \tilde{x}^*(t) + \eta_i(t) = x^*(t) + \delta t + \eta_i(t)$ for $i \in G_1$. Byzantine sensors subtract drift: $y_j(t) = \tilde{x}^*(t) - \delta t + \eta_j(t) = x^*(t) + \eta_j(t)$ for $j \in G_2$.

In both scenarios, the multivariate distribution of $(y_1(t), \ldots, y_N(t))_{t=1}^T$ is identical: group 1 sensors have mean trajectory $x^*(t) + \delta t$, group 2 sensors have mean trajectory $x^*(t)$, and all noise terms are i.i.d. $\mathcal{N}(0, \sigma_\eta^2)$.

Since the observations are statistically indistinguishable, any algorithm that outputs $\hat{\mathcal{B}} = G_1$ is correct in scenario A but incorrect in scenario B (and vice versa). Assigning equal prior probability to both scenarios, the maximum expected accuracy is 1/2.

DT corruption argument. In practice, the digital twin estimates state by fusing sensor inputs. Any robust estimator (median, trimmed mean) applied to a majority-compromised network converges to the Byzantine-controlled trajectory $x^*(t) + \delta t$. Residuals then become:

- Byzantine sensor (in G_1): $r_i(t) = y_i(t) - \hat{x}(t) \approx \eta_i(t)$ — appears honest.
- Honest sensor (in G_2): $r_j(t) = y_j(t) - \hat{x}(t) \approx -\delta t + \eta_j(t)$ — appears Byzantine.

Attribution is inverted: the algorithm labels honest sensors as compromised and vice versa. This behavior is observed empirically in the majority-compromised scenario (scenario S6), where attribution accuracy approaches chance performance (F1 ≈ 0.057), confirming systematic inversion. □

Remark 3. Theorem 2 mirrors the classical result that Byzantine consensus requires $N \geq 3f + 1$ nodes in distributed systems [4]. LA-DT operates under the weaker condition $f < N/2$ (A1) because the digital twin provides a physics-based oracle, effectively substituting for the additional replicas required in pure consensus protocols.

1.7 Extension to Polynomial Drift

Theorem 1 assumes linear drift (A2). The following proposition shows that VGR discrimination extends to polynomial drift of degree $p \geq 1$, with a higher asymptotic coefficient.

Proposition 4 (VGR Under Polynomial Drift). *Let $\Delta y_i(t) = \delta_i \cdot t^p$ for integer $p \geq 1$. Define the polynomial VGR coefficient:*

$$k_p = \frac{\sum_{t=n+1}^{2n} t^{2p}}{\sum_{t=1}^{n} t^{2p}} \xrightarrow{n\to\infty} \frac{\int_{1/2}^{1} u^{2p}\,du}{\int_0^{1/2} u^{2p}\,du} = \frac{1 - 2^{-(2p+1)}}{2^{-(2p+1)}} = 2^{2p+1} - 1. \tag{12}$$

In particular: $k_1 = 7$ (linear), $k_2 = 31$ (quadratic), $k_3 = 127$ (cubic).

In the high-SNR regime ($\delta_i^2 T^{2p} \gg \sigma_\eta^2$), $\mathrm{VGR}_i \to k_p$, and the separation from honest sensors ($\mathrm{VGR} \approx 1$) grows exponentially with drift degree p.

Proof. The continuous integral approximation gives:

$$\frac{\int_{T/2}^{T} t^{2p}\,dt}{\int_0^{T/2} t^{2p}\,dt} = \frac{T^{2p+1}/(2p+1) - (T/2)^{2p+1}/(2p+1)}{(T/2)^{2p+1}/(2p+1)} = \frac{T^{2p+1} - (T/2)^{2p+1}}{(T/2)^{2p+1}} = 2^{2p+1} - 1.$$

For $p = 1$: $2^3 - 1 = 7$. For $p = 2$: $2^5 - 1 = 31$. For $p = 3$: $2^7 - 1 = 127$.

The MSR computation follows the same structure as Lemma 2 with t^{2p} replacing t^2. The separation guarantee of Theorem 1 part (iii) holds a fortiori since $k_p \geq k_1 = 7 > 3$ for all $p \geq 1$. □

Remark 4. The rapidly growing k_p implies that higher-degree polynomial drift is *easier* to detect via VGR, not harder. An adversary seeking to minimize VGR would use the lowest-degree drift ($p = 1$), which is precisely the linear model addressed in Theorem 1. Sub-linear drift ($p < 1$, e.g., $\Delta y_i \propto \sqrt{t}$) reduces k_p below 7 but still yields $k_p > 1$ for all $p > 0$, preserving eventual separation given sufficient observation time.

1.8 GAT Complexity Analysis

Proposition 5 (GAT Per-Layer Complexity). *For a GAT layer with K attention heads, feature dimension F, and input graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $|\mathcal{V}| = N$ and $|\mathcal{E}|$ edges, the computational complexity per forward pass is $O(NKF^2 + |\mathcal{E}|KF)$. For sparse sensor networks where $|\mathcal{E}| = O(N)$ and fixed K, F:*

$$\textit{Complexity} = O(N \cdot F^2 + N \cdot F) = O(N). \tag{13}$$

Proof. Each attention head k computes, for every node i:

1. Linear projection $\mathbf{W}^{(k)}\mathbf{h}_j$ for all $j \in \mathcal{N}(i) \cup \{i\}$: cost $O(F^2)$ per node, $O(NF^2)$ total (each node projects once, shared across heads via batching, or $O(NKF^2)$ without sharing).
2. Attention coefficient $\alpha_{ij}^{(k)}$ computation and softmax over $\mathcal{N}(i)$: cost $O(F)$ per edge, $O(|\mathcal{E}|F)$ per head.
3. Weighted aggregation $\sum_j \alpha_{ij}^{(k)}\mathbf{W}^{(k)}\mathbf{h}_j$: cost $O(F)$ per edge, $O(|\mathcal{E}|F)$ per head.

Summing over K heads: $O(NKF^2 + 2|\mathcal{E}|KF) = O(NKF^2 + |\mathcal{E}|KF)$. With $K = 4$ attention heads and $F = 64$ feature dimensions (as in the main paper GAT architecture), and $|\mathcal{E}| = O(N)$ for spatially sparse graphs: $O(N{\cdot}4{\cdot}64^2 + N{\cdot}4{\cdot}64) = O(N)$.

This linear scaling contrasts with the $O(N^2)$ cost of pairwise feature construction in fully-connected architectures. In experiments with varying network sizes, inference latency increases sub-linearly, confirming the theoretical prediction. □

1.9 Summary of Theoretical Results

Table 1 consolidates the theoretical guarantees established in this appendix.

Table 1. Summary of Theoretical Results

Result	Key Guarantee
Theorem 1	VGR separates honest/Byzantine sensors under linear drift
Theorem 2	Honest majority is necessary for attribution
Prop. 1	DEV diverges to 2 under Byzantine drift
Prop. 2	SCD is positive under Byzantine compromise
Prop. 3	LLR reduces to weighted linear form under exponential family assumptions
Prop. 4	Polynomial drift yields $k_p = 2^{2p+1} - 1$
Prop. 5	GAT layer complexity is $O(N)$ for sparse networks

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